the law's scale:

$$a_0 = \kappa c \sqrt{G\rho_{\mathrm{DE}}}$$

contains c

$\sqrt{G\rho}$ is a rate, like H

$c \times$ rate = acceleration

a halo's scale:

$$a_{\mathrm{halo}} \sim (GM)^{1/3} H^{4/3}$$

no c anywhere

set by mass and expansion

changes with M and with time

near $10^{12} M_\odot$ the two agree only because $c/V_{200} \approx 2000$ happens to be cancelled